AI-Based College Complaint & Resolution System

Mini Project Documentation

Development Platform: Base44 AI

Project Type: B.Tech CSE Mini Project

Live Project: https://campus-care-ai.base44.app

1. Project Overview

The AI-Based College Complaint & Resolution System is a web-based application designed to simplify the process of submitting, managing, tracking, and resolving student complaints. The application uses AI-assisted complaint analysis to help categorize complaints, identify priority, recommend the responsible department, and support faster resolution.

2. Problem Statement

Traditional college complaint systems often depend on manual processes, making it difficult to track complaints, assign them to the appropriate department, monitor progress, and ensure timely resolution. This project provides an intelligent centralized platform to improve transparency, reduce manual effort, and streamline complaint resolution.

3. Objectives

- Provide an easy online platform for students to submit complaints.
- Use AI to analyze and categorize complaints.
- Identify complaint priority and recommend the responsible department.
- Allow staff to manage and update assigned complaints.
- Enable students to track complaint status.
- Provide administrators with dashboards and complaint analytics.
- Improve transparency and reduce complaint resolution time.

4. Key Features

- **Student Module:** Registration, login, complaint submission, ticket tracking, status updates, and feedback.
- **AI Analysis:** Complaint categorization, priority identification, department recommendation, and suggested resolution.
- **Staff Module:** View assigned complaints, update status, add resolution details, and manage department complaints.
- **Admin Module:** Complaint monitoring, user and department management, statistics, and analytics.

- **Notifications:** Status and resolution updates for relevant users.
- **Analytics:** Complaint counts, categories, department performance, and resolution statistics.

5. System Workflow

Step	Process
1	Student submits a complaint
2	AI analyzes the complaint
3	Category, priority, and department are identified
4	Complaint is assigned to the appropriate staff/department
5	Staff reviews and works on the complaint
6	Staff updates the complaint status
7	Complaint is resolved
8	Student verifies the resolution and provides feedback
9	Admin dashboard and analytics are updated

6. User Roles

Role	Main Responsibilities
Student	Submit complaints, track status, receive updates, and provide feedback.
Staff / Department Officer	View assigned complaints, update progress, and resolve complaints.
Admin	Manage users, departments, complaints, analytics, and overall system monitoring.

7. Technologies Used

- **AI Application Builder:** Base44 AI
- **Frontend:** Responsive web interface generated through Base44
- **Backend:** Base44 integrated application backend
- **Database:** Base44 integrated database
- **Authentication:** Role-based user authentication
- **Deployment:** Base44
- **Version Control:** Git and GitHub

8. Advantages

- Reduces manual complaint management.
- Provides centralized complaint tracking.
- Improves department assignment and transparency.
- Helps prioritize important complaints.

- Makes complaint status visible to students.
- Provides useful administrative analytics.
- Demonstrates practical use of AI-assisted web application development.

9. Future Enhancements

- Mobile application for Android and iOS.
- AI chatbot for student support.
- Email, SMS, and push notifications.
- Multilingual complaint submission.
- Image-based complaint analysis.
- Advanced predictive analytics for resolution time.
- Enhanced security, audit logging, and reporting.

10. Conclusion

The AI-Based College Complaint & Resolution System provides a practical and intelligent approach to college complaint management. By combining web application features with AI-assisted analysis, the system helps students submit and track complaints while enabling staff and administrators to manage them efficiently. The project demonstrates how AI-assisted development using Base44 can be applied to solve a real-world college management problem.

11. Live Project

Campus Care AI: https://campus-care-ai.base44.app